

CSE2104 Offline-1

A bookstore maintains a sorted list of unique book IDs to quickly identify whether a book is available in stock. You are required to write a C++ program that uses **Binary Search** to determine if a specific book ID is present in the list. Additionally, the store often needs to find the first book ID that is not smaller than a given value (lower bound) and the first book ID that is greater than a given value (upper bound) in order to manage restocking and categorization. Given a sorted array of book IDs and a query book ID, your program should output whether the book exists in the list, the lower bound position and the upper bound position corresponding to the query.

Tasks

1. Read the number of book IDs and then read the sorted list of unique book IDs.
2. Read the query book ID to be searched.
3. Implement **Binary Search** to check if the book ID exists.
4. Implement logic to find the **lower bound** of the query.
5. Implement logic to find the **upper bound** of the query.
6. Display the search result along with the lower and upper bound indices.

You should write user defined functions for Binary Search, Lower Bound and Upper Bound. Input should be taken inside the main function. All user defined functions must be called from the main function.

Sample Input	Sample Output
7 10 20 30 40 50 60 70 40	Book Found Lower Bound Index: 3 Upper Bound Index: 4

Submit a **.cpp** file and a pdf report ([Sample Report](#)). Follow the naming convention for the .cpp file: **StudentID_Offline1.cpp** and the report: **StudentID_Offline1.pdf**. For example: 180104072_Offline1.cpp and 180104072_Offline1.pdf. Keep the code in the report also.

Follow the submission instructions and deadline strictly.